

22 Game Theory Models

Mapped to the Deal Lifecycle

From landscape assessment to exit: which model activates at each stage of a business decision, what mistake it prevents, what you control, and the takeaway that changes your next move.

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STAGE 1 LANDSCAPE ASSESSMENT

Before committing to anything, read the terrain. These four models answer: "What kind of market am I looking at, and what dynamics are already in motion?"

03

Cournot vs Bertrand

Two identical industries with the same players and costs can produce completely different margins depending on whether firms compete on capacity or on price.

Mistake it prevents	Building a strategy around the wrong competitive dimension, then discovering your margin model assumes a game no one else is playing.
What you control	Your choice of competitive dimension and whether you can shift the market toward differentiation.
Takeaway	Identify whether this market rewards scale or uniqueness before you commit capital to either.

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Beauty Contest

Asset prices in speculative or momentum-driven markets reflect what participants expect other participants to value, which can diverge entirely from what the asset is fundamentally worth.

Mistake it prevents	Running fundamental analysis in a market where price is set by crowd expectations, then being "right" and still losing money.
What you control	Which question you answer first: "what is this worth?" or "what will the market pay?"
Takeaway	In consensus-driven markets, analyze the crowd first, then the asset.

09

Market for Lemons

When buyers in a market cannot distinguish quality, they price for the average, and the best sellers leave first because they are the most underpaid.

Mistake it prevents	Entering a market as a quality seller without a verification mechanism, then being pooled with everyone worse.
What you control	Whether you build or find a quality-signalling channel before you enter, or accept being priced at the market average.
Takeaway	If the market has no way for quality to be observed, either create one or expect to be undervalued.

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Information Cascades

Once two or three participants have moved in the same direction, everyone after them rationally ignores their own private information and follows.

Mistake it prevents

Treating market consensus as independently confirmed evidence when it may trace back to one or two original signals.

What you control

Whether you audit the independent evidence base underneath a consensus before joining it.

Takeaway

Count the independent data points behind any consensus view. If the trail leads back to one or two sources, you are following, not analyzing.

STAGE 2 POSITIONING AND ENTRY

You've read the market. Now establish your position. These four models answer: "How do I enter, signal my value, and read other people's signals?"

04

Entry Deterrence

An incumbent's threat to fight only deters if carrying out the threat is still in the incumbent's interest after entry has already happened.

Mistake it prevents	Being deterred by a threat the incumbent would lose money executing, or making threats you would not follow through on yourself.
What you control	Whether your commitments (or your reading of the incumbent's commitments) are irreversible and visible.
Takeaway	Test every deterrence threat with one question: "After I enter, would they actually spend money to fight me, or would they accommodate?" If accommodate, the threat is empty.

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Spence Signalling

You prove quality to a skeptical market by doing something a lower-quality version of you could not afford to do.

Mistake it prevents	Relying on signals that are equally accessible to worse competitors, which means they carry zero separating information.
What you control	Your choice of signal. The test: "Could someone worse than me send this signal just as easily?"
Takeaway	Every credential, track record item, or upfront cost you cite should be harder for a weaker player to replicate. If it is equally easy, it is decoration.

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Cheap Talk

Any claim that costs the speaker nothing to make carries no information; calibrate incoming claims by what the speaker stands to lose if wrong.

Mistake it prevents	Treating management guidance, pitch deck projections, or letters of intent as meaningful signals without checking what the speaker risks by being wrong.
What you control	Your filter for incoming claims. The question: "What happens to this person if what they just said turns out to be false?"
Takeaway	If the answer is "nothing," discount the claim to near zero. Credibility tracks consequences, not confidence.

Focal Points

When people need to coordinate without explicit communication, they converge on whatever is obvious and conventional, regardless of whether something better exists.

Mistake it prevents	Proposing a technically superior standard or structure that loses adoption to a simpler, more salient alternative.
What you control	Whether your proposed coordination point (standard, convention, default) is intuitive enough that others land on it independently.
Takeaway	If you are trying to establish a standard or a default, simplicity and obviousness beat technical superiority.

STAGE 3 NEGOTIATION AND DEAL STRUCTURE

You are in the room, structuring terms. These four models answer: "How do I divide the value, and what constrains the split?"

02

Nash Bargaining

Your share of any negotiated surplus is determined by the quality of your outside option, and almost nothing else.

Mistake it prevents	Spending 80% of negotiation prep on arguments and 20% on improving your walkaway, when the effective ratio is the reverse.
What you control	Your BATNA and the other side's perception of it.
Takeaway	Before any negotiation, improve your alternative. The strongest argument in the room loses to the strongest outside option.

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Ultimatum Game

People reject offers they perceive as unfair even when accepting would pay them, and this tendency is consistent enough to blow up otherwise profitable deals.

Mistake it prevents	Proposing a split that maximizes surplus extraction without accounting for the counterparty's willingness to reject on fairness grounds.
What you control	The framing and perceived fairness of your offer, independent of its absolute value.
Takeaway	An offer that captures 90% of the surplus and gets rejected is worth less than one that captures 60% and gets accepted. Price in the fairness floor.

05

Chicken

In an escalating standoff, the party that wins is the one who can most credibly eliminate their own ability to back down.

Mistake it prevents	Entering brinkmanship (hostile bid, litigation escalation, regulatory standoff) without a pre-committed position, then yielding to a counterparty who has one.
What you control	Your ability to make your commitment visible and irreversible before escalation peaks.
Takeaway	If you play chicken, commit publicly before the other side does. If you cannot commit first, assess what losing costs and price that into whether you play at all.

Vickrey Auction

Changing the payment rule in an auction (pay the second-highest bid, not your own) flips the optimal strategy from strategic shading to truthful bidding.

Mistake it prevents

Designing a procurement, allocation, or bidding process that rewards gaming over honest value revelation.

What you control

The rules of the process: who pays what, what information is visible, and how the winner is selected.

Takeaway

When you need participants to reveal true valuations, restructure the payment so honesty becomes the self-interested move.

STAGE 4 COMMITMENT AND INVESTMENT

Terms are agreed. Capital is deploying. These four models answer: "How do I commit resources without creating vulnerabilities or overpaying?"

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Winner's Curse

In any competitive bid for an asset of uncertain value, the winner is by definition the bidder who was most optimistic about that value.

Mistake it prevents	Treating an auction win as validation of your valuation, when it is statistical evidence that you were the highest estimator in the room.
What you control	Your pre-auction walk-away price and the discount you apply as bidder count increases.
Takeaway	Set your ceiling before the auction opens. The more bidders, the more you shade down. Tracking "deals lost" is a discipline metric, not a failure metric.

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Hold-Up Problem

Once you have sunk investment into a relationship-specific asset, the counterparty can reopen terms because your exit cost now exceeds the cost of accepting worse conditions.

Mistake it prevents	Front-loading relationship-specific investment without contractual protections, then losing bargaining power because sunk costs give the counterparty leverage.
What you control	The staging of your commitments and the contractual protections you lock in before each tranche of investment.
Takeaway	Every increment of relationship-specific investment should be matched by a contractual protection that preserves your bargaining position. Stage, don't front-load.

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Real Options

The ability to defer, stage, expand, or abandon an investment has quantifiable value that standard NPV analysis sets to zero.

Mistake it prevents	Treating "invest now" vs. "never invest" as the only two choices, when "invest a little now and decide the rest later" is often the highest-value path.
What you control	Whether you structure investments as staged commitments with decision points or as single upfront bets.
Takeaway	In high-uncertainty environments, the option to wait or stage is frequently worth more than the expected return of committing in full today. Structure for optionality.

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Stag Hunt

Both parties prefer the joint outcome and neither is tempted to defect; the only barrier is the fear that the other side will not show up.

Mistake it prevents	Treating a coordination problem as an incentive problem, then trying to change payoffs when the actual fix is credible simultaneous commitment.
What you control	Commitment devices that make both sides' participation visible and simultaneous: escrow, phased matching, public announcements.
Takeaway	When incentives are already aligned but action stalls, the problem is trust. The solution is a structure that removes the option to wait and see.

STAGE 5 ONGOING MANAGEMENT

The investment is live. You are managing operations, relationships, and competition. These four models answer: "How do I sustain cooperation, design incentives, and know when a fight is no longer worth it?"

01

Prisoner's Dilemma

In any repeated relationship, both sides do better cooperating over time, and each side does better cheating in any single round.

Mistake it prevents	Defecting in a relationship that has more rounds left, or cooperating in one that does not.
What you control	Your strategy across rounds (reciprocity, forgiveness thresholds, retaliation signals) and your assessment of how many rounds remain.
Takeaway	The single most important variable is whether the relationship continues. In ongoing relationships, cooperate and retaliate proportionally. In terminal interactions, structure for defection.

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Moral Hazard

When someone's effort is unobservable and their downside is absorbed by someone else, they will rationally take more risk than you want.

Mistake it prevents	Designing a compensation, delegation, or insurance structure that decouples risk-taking from consequences, then being surprised by the risk-taking.
What you control	The incentive structure: co-investment, clawbacks, deductibles, and skin-in-the-game provisions that recouple downside exposure.
Takeaway	Before blaming the agent, audit the contract. If the structure rewards upside and absorbs downside, the behavior you are seeing is the behavior you designed.

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Mechanism Design

People optimize for exactly what you measure, competently and without malice, so the metric you choose is the most consequential design decision you make.

Mistake it prevents	Setting a KPI, comp target, or procurement criterion and being surprised when people optimize for the metric at the expense of the outcome you actually wanted.
What you control	The choice of metric, the reward structure, and the qualitative override layer that catches divergence between the metric and the goal.
Takeaway	Assume intelligent agents. Design for the behavior you want. Build in a manual override for when the metric drifts from the objective.

War of Attrition

Two parties burn resources waiting for the other to quit, and the total cost of the fight often exceeds the value of the prize before either side yields.

Mistake it prevents

Staying in a protracted competition (price war, litigation, subsidized growth race) past the point where cumulative costs have exceeded what winning is worth.

What you control

Your ongoing assessment of cumulative cost vs. remaining prize value, and your willingness to exit early or escalate fast enough to force resolution.

Takeaway

The middle ground (continuing at current burn rate without forcing resolution) is where value gets destroyed. Exit early or escalate decisively. The worst option is staying in the middle.

STAGE 6 EXIT AND RESOLUTION

The position is closing, selling, or unwinding. These two models answer: "What distorts my exit timing, and what can accelerate failure at the end?"

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Prospect Theory

Losses feel roughly twice as painful as equivalent gains feel good, which systematically distorts your timing on when to sell, when to cut, and when to hold.

Mistake it prevents	Holding losing positions to avoid realizing the loss, selling winners too early to lock in the gain, and anchoring to sunk costs that are already irretrievable.
What you control	Your awareness of the framing effect and your willingness to evaluate positions from a clean slate: "Would I enter this today at this price?"
Takeaway	Replace "should I exit?" with "if I held nothing, would I buy this today?" If the answer is no, the loss aversion is making the decision for you.

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Bank Run

A solvent entity fails when enough stakeholders simultaneously demand liquidity, because individual withdrawal is rational even when collective withdrawal is catastrophic.

Mistake it prevents	Underinvesting in liquidity buffers and confidence management because fundamentals are sound, then discovering that solvency does not survive a coordination failure.
What you control	Liquidity reserves, stakeholder communication during stress, and structural features (lock-ups, gates, staggered maturities) that prevent simultaneous withdrawal.
Takeaway	Solvency is necessary and insufficient. Any entity that depends on rolling confidence needs structural defenses against the scenario where everyone rational runs at once.